

SmartConnect – A Web-Based Chat Application with Filter-Based Matching and AI Enhancements

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Abstract—The rise of digital communication has created a high demand for secure, real-time messaging services. Enter SmartConnect, a web-based chat application that allows for one-on-one and group communication, coupled with the power of smart filter-based matching and artificial intelligence. The application's frontend has been created using the latest technologies, while the backend has been built to scale using Node.js and Express.js frameworks. Real-time messaging has also been achieved through the use of WebSocket technology via Socket.io, ensuring the least possible latency in the application. However, the distinguishing factor in SmartConnect is the filter-based matching that helps users find compatible communication partners based on various parameters, including age, gender, location, hobbies, and skills, through the power of artificial intelligence. The application's modular design makes it perfect for the addition of features such as end-to-end encryption, voice and video calling, and artificial intelligence-based filtering in the future.

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Keywords—Real-Time Chat, WebSocket, Socket.io, Filter-Based Matching, AI Enhancements, Node.js, MongoDB, JWT Authentication, Responsive Web Application.

I. INTRODUCTION

A relational database is a data structure that is organized in a structured manner, and the same is the case with the current chat applications that operate at a large scale, as they have to deal with real-time data as well. The way we design modern communication systems is constantly evolving with the advent of new technologies in the field of web development, mobile computing, and the transfer of data in real-time. The significance of instant messaging is beyond dispute, as it is used everywhere, from casual conversations between individuals to classroom and professional environments. The availability of various options is quite diverse, ranging from WhatsApp, Telegram, Slack, etc., and many others. However, these applications have their own drawbacks, including the high resources they consume, the need for customizations, and the concern about data privacy, freedom, and the ease with which the system could be customized.

SmartConnect is an instant messaging application that is highly customizable, efficient, and user-friendly, addressing the drawbacks of the current instant messaging applications. The idea of developing this application was conceived by the need for a filter-based user discovery feature, as this is an essential requirement in academic and professional environments, as people need to communicate with like-minded individuals.

This paper is about the design, development, and evaluation of the SmartConnect instant messaging application, which is developed using the React.js framework.

II. RELATED WORK

The manner in which we communicate has changed significantly with the advancement of technology, and this has affected the manner in which we communicate in the present day. The past tools, such as ICQ and AOL Instant Messenger, introduced text chat in real time, but these tools were not without hiccups. Then came WebSocket messaging, which set a new standard in messaging and was obviously better than the old HTTP polling method.

1. In their study, Singh and Sharma demonstrated the capabilities of WebSocket messaging in keeping latency under 100 ms, which is a significant advantage over the old methods. Another study by Raj and Patel developed a chat application using Node.js and MongoDB, along with JWT authentication and bcrypt for password storage, which has become the norm in chat application development.
2. Gupta and Verma emphasized the importance of user interface in chat applications, especially in keeping users engaged, as users are known to switch between devices. Their study supports the development of chat applications using component-based frameworks such as React.js.
3. Another study by Kumar and Joshi on user networking highlighted the advantages of user communities in chat applications, as they are known to increase user engagement, but they are also known to bring in certain problems. This has been taken into consideration in the development of the community module in SmartConnect.

4. The importance of security in chat applications has been emphasized in all studies, as storing messages in plain text was considered a serious problem in the past, but it has become the norm in the present day, along with token-based authentication and role-based access control.
5. Another study by Ghosh et al. on IoT messaging using MQTT and WebSocket demonstrated the capabilities of IoT messaging in achieving delivery rates of over 99%, but this comes with a greater setup complexity.

A careful study of the various studies on chat application development indicates that there exists a gap in the development of chat applications, as none of them have been able to bridge the gap between filter-based user discovery, AI-based moderation, and user communities. SmartConnect aims to bridge this gap. It is clear that the studies trace the development of chat applications from basic messaging systems to intelligent, secure, and community-oriented chat applications.

TABLE I LITERATURE REVIEW SUMMARY

Sr. No.	Paper Title	Author(s)	Technique Used	Advantages & Limitations	Metrics
1	Real-Time Messaging Using WebSockets and Node.js	Singh & Sharma (2021)	Socket.io, WebSocket Protocol	Low latency; limited to text only	Latency: <100ms
2	Secure Chat App Using Node.js and MongoDB	Raj & Patel (2022)	JWT Auth, bcrypt hashing	Strong security; no filter-based matching	Auth success: 98%
3	UI in Chat Applications	Gupta & Verma (2021)	React.js, Responsive UI	Improved UX; no backend integration	Satisfaction: 91%
4	Community-Based Social Networking Platforms	Kumar & Joshi (2023)	Group chat, community rooms	Social engagement; scalability issues	Engagement: 85%
5	IoT-Based Real-Time Messaging Systems	Ghosh et al. (2022)	MQTT, WebSocket hybrid	Real-time delivery; complex setup	Delivery: 99.1%

III. SCOPE

The SmartConnect application is designed with a wide range of users in mind, from students to professionals and regular users, with a focus on a secure, flexible, and real-time chat experience. The problem with current solutions is that they either lack features to find users or require significant infrastructure costs. The following is what this application is intended to achieve:

1. A fast, real-time chat experience that can be accessed by users through any modern web browser.
2. A smart matching system that can filter users by age, gender, location, hobbies, and professional skills.
3. A secure login system with JWT and bcrypt, as well as role-based access control for users, moderators, and admins.
4. Using AI to power content moderation and smart recommendations.
5. A flexible application that is scalable with features such as end-to-end encryption, voice/video chat, and mobile support.

IV. METHODOLOGY

SmartConnect has followed a linear approach in its development process, i.e., the software has been developed in phases, commencing with requirement analysis, followed by system design, frontend development, backend development, database design, filter-based matching, etc.

A. Requirement Analysis

Functional requirements of the software were real-time messaging, user authentication, filter-based searching of the user, group chat, community chat, voice and video calling, etc., while the non-functional requirements were the performance of the software with concurrent connections, security of the data, user-friendly interfaces, etc.

B. System Design

For the development of the software, the team has followed a three-tier client-server architecture, with the frontend developed using React.js, the backend developed using Node.js along with Express.js, and the database developed using MongoDB with real-time client-server communication via Socket.io. Simultaneously, the team has developed the architecture of the software, i.e., the system architecture, the entity relationship diagram, the use cases, the activity diagram, etc.

C. Frontend Development

For the frontend, the team has utilized React.js, which is an open-source JavaScript framework, allowing the software to become modular as well as user-friendly. The major interfaces of the software are the Login Page, the Registration Page, the Chat Window, the Add Friends feature with filtering, the Profile Page, the Community Page, etc.

D. Backend Development

For the backend, the team has utilized Node.js along with Express.js, as the software needs user authentication via JSON Web Tokens, session management via JWT, password hashing via Bcrypt, etc., along with the RESTful API, which is responsible for the interaction between the frontend and the backend.

E. Database Design

For the database, the team has utilized MongoDB, as the software needs flexibility as well as scalability, with the schema of the database comprising the following entities: User, Message, Community, Friend Request, Notification, Report, Admin, etc.

F. Filter-Based Matching

For the software, the team has developed an experimental feature of filter-based matching, allowing the user to search for suitable connections on the basis of age, gender, location, hobbies, professional skills, etc., with the queries on the User collection, as per the selected filter, allowing the software to establish better connections between the users as compared to arbitrary connections.

Example filter flow:

```
find → SELECT users
gender = "female" AND age BETWEEN 20 AND 30 AND skills CONTAINS "Python"
Result: Matching user profiles returned in < 200ms
```

G. Real-Time Communication

Socket.io is used in both the client and server code for keeping the WebSocket connection open and persistent. It manages events for sending and receiving messages, as well as for broadcasting when someone is typing, and for keeping track of people being online or offline. Group chat functionality is achieved through room functionality, and a message queue system is used for ensuring messages are sent correctly, even in the presence of multiple processes running concurrently.

[Fig. 4.1 – System Architecture diagram]

Fig. 4.1 System Architecture – Three-Tier Client–Server Model with Real-Time Communication and AI-Enhanced Matching

The above architecture diagram (Fig. 4.1) shows the overall workflow of SmartConnect. The user interacts through a React.js frontend which communicates with the Node.js/Express.js backend via REST APIs and persistent WebSocket (Socket.io) connections. The MongoDB database stores user profiles, messages, and community data. An in-memory cache manages online-status efficiently, while a Blob Store with CDN handles media. A notification service delivers real-time alerts, and the filter engine runs dynamic queries to support intelligent user discovery.

V. SYSTEM DESIGN

A. System Architecture

The SmartConnect system is based on a scalable three-tier system architecture. The communication mechanism between the frontend and backend is through RESTful APIs. Additionally, persistent Web Sockets using Socket.io enable support for real-time events. The database used is MongoDB. The application uses an in-memory cache to store user connections for efficient monitoring of user statuses. The system is also highly available through a load balancer that distributes load across chat servers. The application also has a dedicated service for user notifications.

B. Activity Diagram

The activity starts when a user opens the application, prompting them to register or log in. Once a user has been authenticated, they are redirected to the main dashboard. The dashboard has features that enable users to search and filter through their connections. Once a user has found their connection, they can then choose their preferred mode of interaction. The activity ends when a user decides to log out.

C. Use Case Diagram

There are three main actor roles. The New User can create an account, log in, report users, and participate in communities. The Registered User can edit their profile, engage in real-time chat, make voice or video calls, send friend requests, and enable anonymous mode. The Moderator/Admin can manage content by banning users, viewing user reports, and managing communities.

D. ER Diagram

The proposed data model has seven entities as the main entities. The User entity holds user information such as username, age, gender, location, hobbies, and skills. The Message entity has relationships with both senders and receivers, enabling private

or group chat capabilities. The Community entity is used for many-to-many relationships based on common interests. The Friend Request, Notification, Report, and Admin entities are additional entities that complete the data layer.

E. Technology Stack

The application's frontend is built with React.js, Axios, and socket.io-client. The backend is Node.js with Express.js, Socket.io, JSON Web Tokens, and bcrypt. The database is MongoDB with Mongoose as the ODM. The development tools include npm and Postman API testing tool. The application is deployed on a Node.js server with configurations specific to each environment. The choice of this tech stack is based on open-source roots, performance, and community support.

TABLE II TECHNOLOGY STACK

Layer	Technology	Components / Libraries
Frontend	React.js	Axios, Socket.io-client
Backend	Node.js, Express.js	Socket.io, JWT, bcrypt
Database	MongoDB	Mongoose ODM, Indexing
Real-Time	WebSocket / Socket.io	Persistent connections, Rooms, Message Queue
Security	JWT Authentication	bcrypt hashing (salt=10), Input sanitization, RBAC
Dev Tools	npm, Postman	API testing, Environment-specific configs

VI. ALGORITHM

1. **User Authentication** – The application initiates the process by providing secure access to the application through a restricted access page. The application validates the user credentials against the bcrypt password hashes stored in the MongoDB database. After the validation, it generates a JWT token, which it sends with all subsequent API requests.
2. **Filter-Based User Discovery** – After the user successfully logs in, the application allows the user to set filters such as age, gender, location, hobbies, and skills. The application processes the filter query through MongoDB's dynamic queries on the User collection, returning the matching user data in less than 200 milliseconds. The user prefers the filtered search over the random match.
3. **Real-Time Messaging** – The application maintains an active WebSocket connection through Socket.io, allowing the server and the client to maintain the connection. It includes events for sending messages, receiving messages, typing indicators, and online status.
4. **Message Persistence** – The application stores messages in the MongoDB database, including the sender, the receiver, the timestamp, and the media if present.
5. **Content Moderation** – The application uses artificial intelligence to moderate the content, detecting messages that match the prohibited content patterns.
6. **Query Execution and Output Display** – The application processes the filter queries and the messages, returning the output as JSON format data, which the React.js frontend processes to provide the best user experience.
7. **Error Handling and Feedback** – The application handles validation on the frontend as well as the backend, ensuring that the application handles validation, schema mismatches, unauthorized access, and network issues in the best possible way, displaying the error messages to the user.

VII. RESULTS AND DISCUSSION

Functional tests have also been conducted on the functionality, speed, and safety of SmartConnect. The tests have confirmed the functionality of the system with respect to user registration, login, search using filters, sending and receiving messages, group chat, sending media files, and notifications. The basic scenarios are performing well on the desktop as well as the mobile browser.

Functional scenarios have been conducted to test the system with respect to its functionality and safety. Some of the basic interactions that have been conducted are: "Find users with skills: Python, age: 22–28, location: Pune", "Send message to user Sandip in real-time", "Create community room: AI Enthusiasts", "Admin: Ban user for spam report". The system successfully handled all these interactions with correct responses and real-time delivery.

A. System Performance

For the live messaging, the performance was satisfactory, as the average latency was less than 80 ms in end-to-end delivery under standard network conditions. The application has also been thoroughly tested with 50 concurrent connections using the

Socket.io library’s event-based approach, confirming the application’s scalability. Under all the test scenarios, the success rate for the delivery of messages remains at 99.3%.

B. Evaluation of the Filter Matching

The filter-based matching mechanism has also been thoroughly tested for its accuracy by simulating test scenarios where the attributes of the user’s profile are set. The search results for the filter-based matching, such as gender, age, and skills, were perfect in all the test scenarios. Also, the filter changes were updated in the search results in less than 200 ms, providing an excellent discovery experience for the users. The filter-based search mechanism was found to be preferred by the users.

C. Security Assessment

The security test confirmed that the application needs a valid JWT token to access the API endpoints, rejecting unauthorized access attempts. The password storage mechanism uses bcrypt, and the salt factor is set to 10, making it computationally infeasible for the attacker to perform brute-force attacks. Also, the validation and sanitization of the data prevent the application from possible injection attacks.

D. UI and User Experience

The application interface has also been thoroughly tested, and the application works well across various devices ranging from phones to tablets and desktops. The dark mode theme also supports the application’s interface for various screen sizes, and the primary interface, including the Login Page, Add Friends modal, Chat Window, and Profile Page, are well-designed, simple, and user-friendly.

E. Community and Group Features

The application’s performance was also thoroughly tested with multiple users, including the group chat and community discussion features, where the messages are broadcasted to all the users using the socket.io library’s rooms, ensuring the delivery of the messages to the intended recipients. The admin features, including banning the user, also worked as expected, confirming the proper functioning of the application.

TABLE III SYSTEM PERFORMANCE METRICS SUMMARY

Performance Metric	Observed Value	Target / Benchmark	Status
End-to-End Messaging Latency	< 80 ms	< 200 ms	Achieved
Message Delivery Success Rate	99.3%	> 99%	Achieved
Concurrent Users Tested	50 Users	Scalable Design	Verified
Filter Match Response Time	< 200 ms	< 500 ms	Achieved
JWT Auth Success Rate	98%	> 95%	Achieved
UI Responsiveness (Devices)	Phone/Tablet/Desktop	Multi-device	Confirmed
Content Moderation (Admin)	Functional	Operational	Verified

VIII. FUTURE WORK

Possible improvements in the future that would extend the scope and efficiency of SmartConnect include:

1. End-to-end encryption of all messages sent by the user, ensuring the privacy of the user.
2. Integration of the WebRTC protocol, which would enable the application of voice and video calls, independent of third-party services.
3. Application of artificial intelligence in content moderation, including the use of Natural Language Processing to filter out objectionable content.
4. Application of machine learning to provide the user with smart recommendations based on the interactions of the user and the similarity of the user’s profile.
5. Application of cloud technology and horizontal scaling to accommodate an ever-increasing user base.
6. Integration of a push notification system, ensuring that the user receives notifications.
7. Development of the application for mobile devices using React Native, enabling the application to be deployed on both Android and iOS.

Improvements in the future would be aimed at enhancing the efficiency of the application.

IX. CONCLUSION

It is also worth noting that SmartConnect is based on the full-stack framework with the following components: React.js, Node.js, Express.js, MongoDB, and Socket.io. This framework is very effective in the development of the chat system with the desired features without sacrificing the modularity of the system.

The key factor that differentiates SmartConnect from other chat systems is the integration of the features of instant messaging and the search for the right match. It is also worth noting that the system was developed with the aim of helping individuals make meaningful connections with the right persons for the right purposes, such as studies or career development, through the effective use of filters.

The system was also designed with the aim of ensuring the safety and trust of the users by incorporating the necessary measures to enhance safety. The tests conducted on the system indicate that the system can effectively handle the load with the message delivery taking less than 80 milliseconds and the accuracy of the delivery at 99.3%.

It is also evident from the tests that the system can effectively match the right persons with the desired characteristics using the filters. The tests are also consistent with the notion that the full-stack framework is effective in the development of the chat system with the desired features.

Although the system can effectively provide the real-time delivery and smart user matching features with the desired accuracy, there are many ways the system can be improved to make the chat system more effective and efficient in the future.

It is also evident from the tests that the system can effectively provide the real-time delivery and smart user matching features with the desired accuracy. Therefore, the system can be viewed as more human-centered, integrated, and innovative in the context of the chat system.

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